KaxaNuk
Sharing knowledge

— INVESTMENT BOOTCAMP

# GETTING THE DATA AND FEATURE ENGINEERING.



**03**

From your provider to an analyzed signal: the Data Curator, the Data Refinery and the Data Analyzer, with Golden-Flow beside you. Step 3 of the eight.

IN THIS SESSION


01 GET

02 REFINE

03 ANALYZE

04 WRAP-UP

# FROM THE DATA TO AN ANALYZED SIGNAL.



## 1 GET

Your data, from your provider.

- Data Curator: four providers
- Your whole universe, in one download
- Your c_* features: trend and traded value

**You leave with: your data**

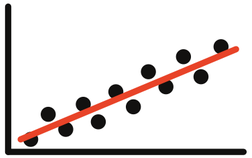


## 2 REFINE

Your universe, date by date.

- Data Refinery: reads it all
- Ranks and z-scores across stocks
- Your weight: the share of traded value

**You leave with: your panel**



## 3 ANALYZE

A signal, or just noise?

- Data Analyzer: explore your signal
- Against buy and hold
- The same strategy, another signal

**You leave with: keep or drop**



## 4 WRAP-UP

What you learned, and what comes next.

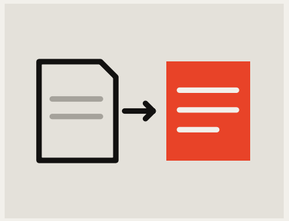
- The session's ideas, in your words
- Homework: finish your analysis
- Next: your portfolio, in S04

**You leave with: your plan for S04**

# EIGHT STEPS. IN THIS SESSION, THE THIRD.

S02 | THIS SESSION | S04 AND S05 | S06 | BEYOND THE COURSE

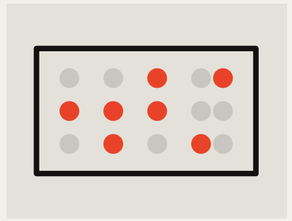
1 2 3 4 5 6 7 8



### RESEARCH

S02

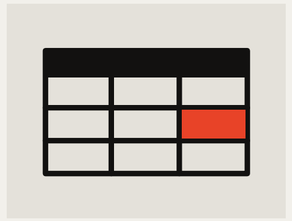
Your objective, from your library · OBJECTIVE.md



### UNIVERSE

S02

Which companies and when; delisted included · Universe/

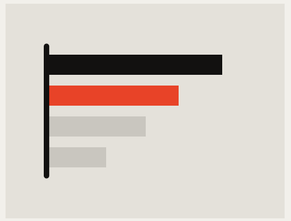


### DATA

S03

Your features, and whether they carry a signal · Data/

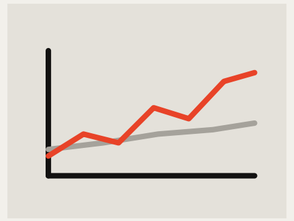
STEP 3



### PORTFOLIO

S04

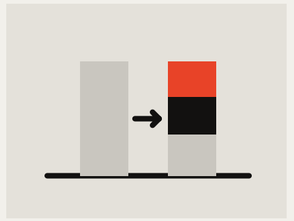
Your hypothesis, then the portfolios · Experiments/



### BACKTEST

S05

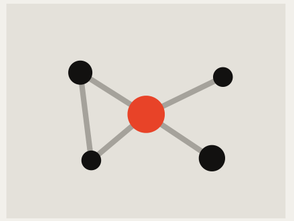
Your portfolio through time, net of costs



### ATTRIBUTION

S05

Where the return came from



### PAPER TRADING

S06

Tested without real money · Paper_Trading/



### PRODUCTION

outside the course

**From S02's homework**

your chosen provider, its key in `Config/.env`, and your seed with the dead stocks.

# THREE LIBRARIES, ONE STEP: YOUR DATA AND FEATURES.

| CURATOR | REFINERY | ANALYZER |
| --- | --- | --- |
| **DATA CURATOR**<br>Downloads, validates and harmonizes data from four providers; your own calculations per instrument.<br>**Get. Validate. Unify.** | **DATA REFINERY**<br>Reads it all and builds features across stocks and through time: ranks, z-scores, sector neutral.<br>**Enrich. Combine. Scale.** | **DATA ANALYZER**<br>Explores your features and your signals: EDA, IC, charts and a simple backtest.<br>**Explore. Detect. Understand.** |

**Feature engineering**
from raw data to a signal worth testing, before any portfolio.

# ONE CODE, ANY PROVIDER.

| WHAT IT DOES | WHY IT MATTERS |
|---|---|
| **Downloads from four providers: LSEG, FMP, Yahoo Finance and Sharadar.** | You use the one you have. |
| **The same tags for all: `m_`, `f_`, `fbs_`, `fis_`, `fcf_`, `d_`, `s_`.** | Already mapped to each provider: you switch without touching your code. |
| **Your own `c_*` calculations, over each instrument's history.** | Simple features: moving averages, trend, traded value. |
| **And `r_*` for the Data Refinery's calculations.** | You know where each column comes from: `c_*` from the Data Curator, `r_*` from the Data Refinery. |
| **Validates and harmonizes what arrives.** | Catches a provider's outliers before they reach a signal. |
| **One place for each API.** | If a provider changes its API, it is updated once and everything works again: strategies, material, courses. |
| **Open source.** | It can grow: FRED, KaxaNuk's Analytics, Bloomberg, Intrinio, Databento, or your own databases. |

**LESS MAINTENANCE** your code depends on the Data Curator, not on each provider's API.

# USE THE ONE YOU HAVE. KNOW ITS GAPS.

**Your provider**

Each with its own identifier.

### Yahoo Finance

**No fundamentals, no dead stocks.** To start; limited in a backtest.

no key

### FMP

Fundamentals, dividends, splits.

a key · Basic, free

### LSEG Workspace

VWAP; quarterly fundamentals only.

a subscription

### Sharadar

Matches KaxaNuk's Analytics: the one in the course's slides.

paid

Your seed uses your provider's `main_identifier`: if you switch providers, check it.

# ONE DOWNLOAD, YOUR WHOLE UNIVERSE.

ARYA · TREND-AND-VOLUME

Download my whole universe, the dead stocks too. 1

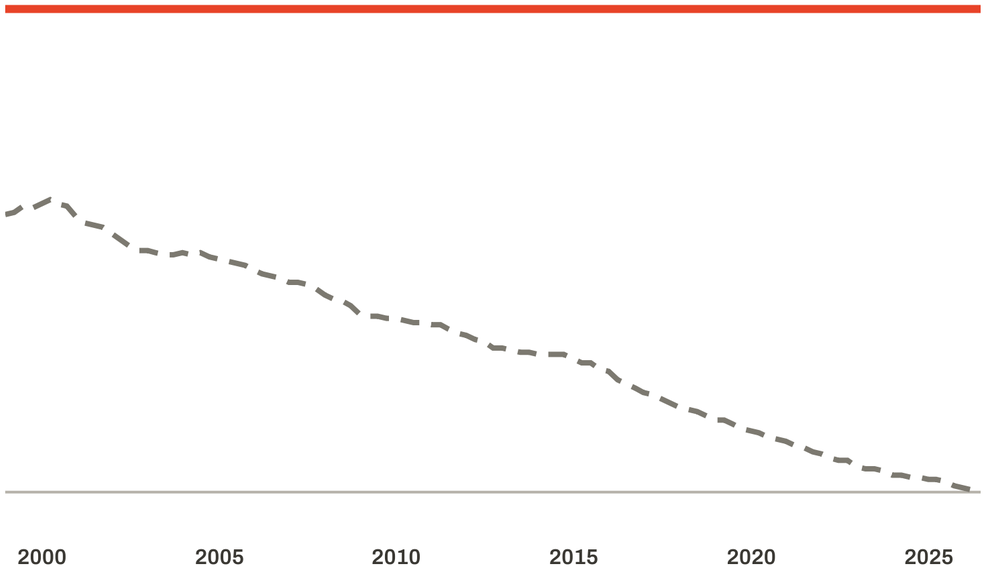
With your seed and your provider, two passes: the second, only what was missed. Go ahead?

go ahead

Done: one file per stock. Now, the coverage. 2

COVERAGE · KN US EQUITY CORE · EVERY REVIEW

Sharadar · KN US Equity Core · provisional




Every member of every review has a price. Of the 643 in 1998, more than half no longer trade: without them, your universe lies.

1. Your whole universe, the dead stocks too.
2. Then the coverage: from when it is usable.

# YOUR FIRST FEATURES, PER INSTRUMENT.

DATA/CURATOR/

custom_calculations.py 1

| | |
|---|---|
| c_sma_50 | 50-day average |
| c_sma_200 | 200-day average |
| c_trend_50_200 | 50 against 200 |
| c_golden_cross | 1 or 0: the cross |
| c_traded_value | VWAP × volume |
| c_traded_value_sma_63d | its 63-day average |

ARYA · TREND-AND-VOLUME

Add the 50- and 200-day averages, their cross and the traded value. 2

I check the built-ins first. I propose six c_*, on the adjusted price. Go ahead?

go ahead

Written. Run the curator again: every stock. 3

1. c_*: one instrument's history.
2. Reuse first: the Features page.
3. A variant, like 63/252, is another c_*: another download.

`data-curator-custom-calculations`

# FROM EACH STOCK TO YOUR WHOLE UNIVERSE.

| WHAT IT DOES | AN EXAMPLE |
|---|---|
| **Reads everything you downloaded** | The Data Curator's files, and those of any API it does not have yet. |
| **Compares across stocks, on each date** | A rank or a z-score of your signal against the whole universe. |
| **Sector neutral** | Point-in-time classifications, to compare within each sector. |
| **Weights among the eligible** | Each stock's traded value over the total of its date's eligible stocks: Golden-Flow's weight. |
| **Factor and signal metrics** | Any calculation that needs the whole universe: N stocks by N. |

**BY DATE** a rank is taken within its date; a value dated t reads up to t – 1.

# YOUR WEIGHT, AMONG THEM ALL.

**DATA/REFINERY/**

`custom_calculations.py` 1

- `r_traded_value_share` its part, by date
  - `c_golden_cross` who is eligible
  - `c_traded_value_sma_63d` how much it trades

`Time_Series/` the panel

**ARYA · TREND-AND-VOLUME**

The share of traded value, among each date's eligible stocks. 2

Plan: by date, only with the cross at 1, summing to 1. Go ahead?

go ahead

It ran: `Data/Refinery/Time_Series/`. 3

1. Golden-Flow: its weight, in the refinery.
2. It needs every stock of the date: `r_*`.
3. The diff: by date? Does it sum to 1?

# A SIGNAL, OR JUST NOISE?

| WHAT IT DOES | WHAT FOR |
|---|---|
| **Exploratory data analysis (EDA)** | Distributions, outliers, coverage and stability: understand your data before using it. |
| **Ready-made analyses and charts** | IC, ICIR, ranks and deciles, on the platform or in code. |
| **Statistics and machine learning** | Several methods to measure whether a signal anticipates returns. |
| **A simple backtest** | The signal alone, with the Backtest Engine, before mixing it into the portfolio. |
| **Your own notebooks** | marimo or Jupyter: analyses to your measure, with comments. |
| **Findings for the team** | Documented and commented, to discuss them fast. |

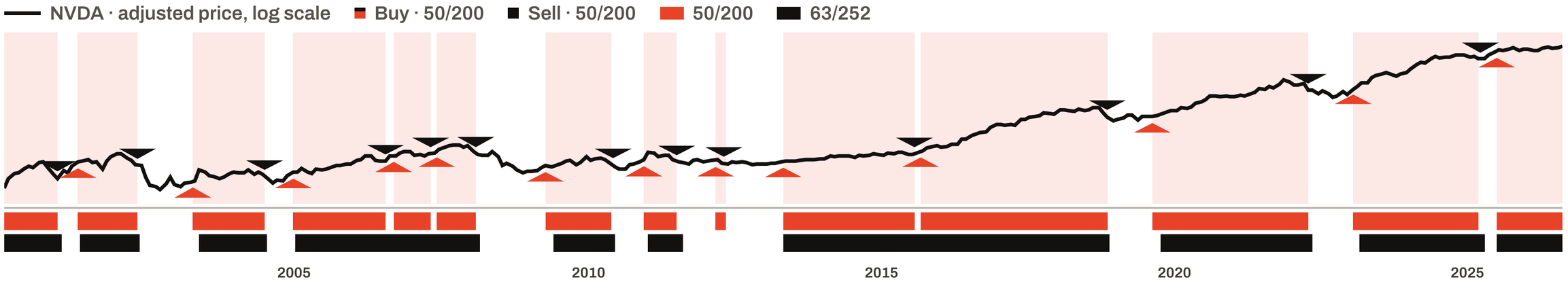
**BEFORE THE PORTFOLIO** a signal earns its place alone, or is dropped.

# EXPLORE YOUR SIGNAL, STOCK BY STOCK.

SIGNAL EXPLORER · NVDA · 50/200 AND 63/252

Sharadar · to 2026-08-14

Stock: NVDA ▾ | Signals: 50/200, 63/252 ▾ | Log scale




Price adjusted for splits and dividends, at each month's close; each strip, the days its signal is at 1. From 2000-01-20 to 2026-08-14.

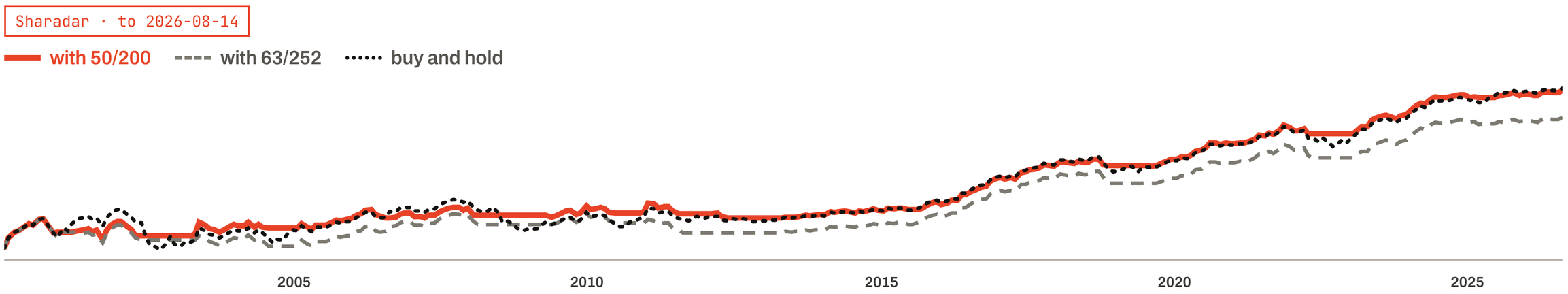
**1 and 0**
buy when it turns on, sell when it turns off; before 252 days, no signal.

**One stock teaches**
the rule; it does not decide. NVDA is a known winner.

# HOW MUCH DID YOUR SIGNAL MAKE?

NVDA · GROWTH OF 1 · 50/200, 63/252 AND BUY AND HOLD

Sharadar · to 2026-08-14




| MAX DRAWDOWN · 50/200 | MAX DRAWDOWN · 63/252 | MAX DRAWDOWN · HOLD |
|---|---|---|
| -70.0% | -76.8% | -89.8% |

Log scale. Buy when it turns on, sell when it turns off, one day later, no costs. 1 became 2,323, 610 and 2,618.

**The 50/200**
came close to holding, with a far smaller drawdown.

**The 63/252**
came late to the rallies and fell behind.

# 50/200 OR 63/252? ONE STOCK AND TWENTY.

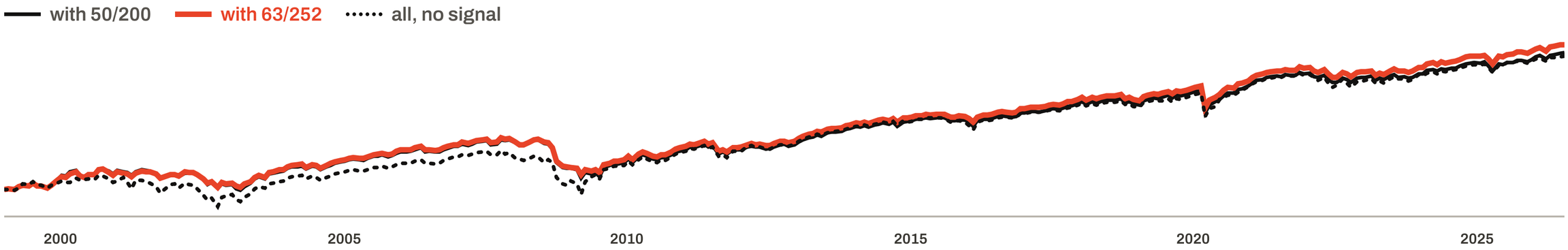
| PER STOCK | 50/200 | 63/252 | BUY AND HOLD |
|---|---|---|---|
| **NVDA · Sharpe** | 0.87 | 0.74 | 0.80 |
| **NVDA · max drawdown** | −70.0% | −76.8% | −89.8% |
| **Top 20 · best Sharpe** | 5 of 20 | 3 of 20 | 12 of 20 |
| **Top 20 · median Sharpe** | 0.57 | 0.53 | 0.56 |
| **Top 20 · median drawdown** | −62.9% | −56.6% | −73.5% |

**AMONG THE LARGE** holding usually wins on Sharpe; the signals cut the drawdown.

# THE SAME STRATEGY, ANOTHER SIGNAL.

EQUAL WEIGHT ON THE CROSSED-ON · EVERY 21 DAYS · KN US EQUITY CORE

Sharadar · KN US Equity Core · provisional




| SHARPE · 50/200 | SHARPE · 63/252 | SHARPE · ALL |
|---|---|---|
| 0.62 | 0.67 | 0.54 |

Growth of 1, log scale, no costs; each date's members, the dead included.

**Change one thing**

same universe, weight and rebalancing; only the signal changes.

**In a portfolio it helped**

more Sharpe and less drawdown than all with no signal.

# YOUR SIGNAL CANNOT KNOW THE FUTURE.

**Look-ahead**

Using what you did not know yet.

**The price**

Today's signal uses prices up to yesterday.

**The report**

An earnings figure counts from its release.

**The adjustment**

Only data that existed on that date.

**The assistant**

It knows what happened: figures from your files.

The same delay for every signal you compare.

# WHAT WORKS BEST ACROSS ALL STOCKS?

1,919 KN US EQUITY CORE STOCKS · AVERAGE · SHARADAR

| AVERAGE | 50/200 | 63/252 | HOLD | |
|---|---|---|---|---|
| Annual return | 1.7% | 2.0% | -0.6% | |
| Sharpe | 0.17 | 0.19 | 0.25 | |
| Sortino | 0.29 | 0.32 | 0.39 | |
| Calmar | 0.08 | 0.09 | 0.07 | |
| Max drawdown | -51.8% | -51.8% | -67.8% | |
| The 713 that fell | -7.4% | -6.9% | -23.8% | 1 |
| The 1,206 that rose | 7.0% | 7.4% | 13.1% | 2 |

1. Those that fell: the signal cut the loss to a third.
2. Those that rose: it cost half the return.

# YOUR ANALYSIS, IN WRITING.

ARYA · TREND-AND-VOLUME

Compare 50/200 and 63/252 against holding, across my whole universe. 1

Explorer, a table per stock and the same strategy with each signal. Go ahead?

go ahead

Done, in RESULTS.md. Your decision is missing, with N = 2. 2

RESULTS.MD · EXAMPLE: THE KN US EQUITY CORE

| SIGNAL | SHARPE | MAX DD | AGAINST ALL | DECISION |
|---|---|---|---|---|
| 50/200 | 0.62 | -50.9% | +0.08 | |
| 63/252 | 0.67 | -49.1% | +0.12 | |
| All, no signal | 0.54 | -55.4% | — | |

The decision is yours: against the threshold you saved before looking; N = 2.

1. Ask for the full analysis.
2. Decide and write it, against your threshold.

`data-analyzer-runs`

# THE SESSION'S IDEAS, IN YOUR WORDS.

**01**

Why does one code serve any provider?

**02**

When is a feature `c_*`, and when `r_*`?

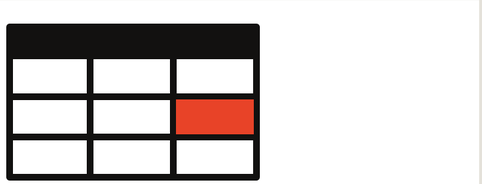
**03**

Why can a signal that makes less than holding still be useful?

**04**

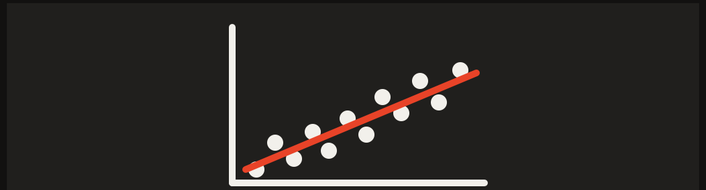
Why analyze a signal alone, before the portfolio?

# BEFORE S04: ARRIVE READY.



## 01 YOUR DATA

- Your whole universe, the dead too.
- Check the coverage, and from when.
- Your `c_*`, with its report.
- Ask `next`.



## 02 YOUR SIGNAL

- Your `r_*`: your weight, by date.
- Explore it; compare it with holding.
- Its table, in the Analyzer.
- Keep or drop, in `RESULTS.md`.



## 03 READY FOR S04

- Your kept signal, with its table and N.
- Golden-Flow: its `RESULTS.md`.
- Golden-Flow: read its `BLUEPRINT_1.md`.
- How you would weight.

# ANALYZED FIRST. BUILT NEXT.

Your data, your features and your signal, analyzed before any portfolio.

**LEARN**

The Bootcamp continues, then the pillars.

`github.com/KaxaNuk/Investment-Bootcamp`

**TEST**

Your strategy on the Lab; universes and factors from Analytics.

`kaxanuk.mx/lab` · `kaxanuk.mx/analytics`

**TELL US**

What your researcher helped you do.

QUESTIONS `workshops@kaxanuk.mx`

NEXT

## Session 04 · Write Your Hypothesis and Build the Historical Portfolios

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Nothing here is a recommendation to buy, sell or hold any security; backtests, Golden-Flow's included, are hypothetical, in sample and never traded, and their analysis is a measurement of past data, not a result; illustrative charts carry no data. Your researcher can be wrong; every decision is yours. What your researcher reads goes to the company behind your assistant, never to KaxaNuk.

Data providers' pages, FMP's plans, tools and works read 2026-10-11; tools checked again 2026-10-13; the figures, from Sharadar prices of the KN US Equity Core, whose methodology is still provisional, computed 2026-10-17, one day late and with no costs. Shared under CC BY 4.0; the KaxaNuk name, logo and isotipo are not.